SEACITY UNIVERSITY

Faculty Design, Orbital Mechanics and Aviation Department Science

Master's Thesis
zur Erlangung des akademischen Grades M.A.

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CommNet Kerbin System

A web of three relay satellites per body to ensure a full coverage of the kerbin system

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temporary version from September 12, 2020

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1 Kerbin Satellites

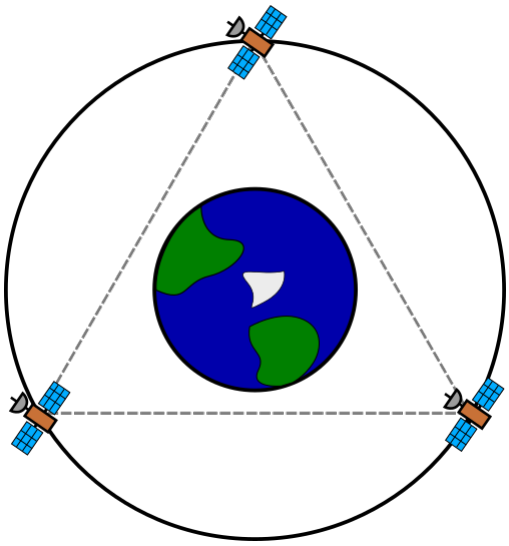


Figure 1.1: Kerbin Relay Network

We want to create a communications network with three equally distanced satellites in an equatorial orbit around planet kerbin. There is a simple reason for this:

For the controll of a spaceship, a space station or a satellite we usually need an active connection to the kerbal space center (KSC). But by positioning three equally distanced satellites in the same orbit around the planet we can prevent that the connection get's cut off as soon as the KSC disappeares behind the horizon. Algorithms will automaticly choose the shortest connection over as few satellites as possible.

Kerbin is a relatively small planet. Kerbins 600km radius measures just one tenth of the radius of Planet Earth. Also the game won't calculate transmission time delays (Latency), which is why we only need to ensure that the connection holds, not how good exactly the connection actually is. Another point is that we want enable an interplanetary communicatin with our CommNet. Because of that we want to choose an antenna strength that automaticly must be stronger than what we would need for a communication between our three satellites. In order to

survey as much from the round planets surface as possible (just think about the difficult to reach poles), we will set the satellites into a high orbit. The further away the observer from a sphere, the larger the visible surface gets.

However, if we wanted to create an internet network, the way SpaceX does it with Starlink, then a lower orbit would be much more favourable in order to shorten the latency. But this would require much more satellites per orbit, as the communication would otherwise have to go through the planets surface, which isn't possible after all. The satellites must be at least high enough to see each other just above the horizon.

We choose an equatorial orbit, mainly because it will require less fuel and therefore less weight for the rocket. Also the kerbal space center (KSC) lies directly at the equator, hence one must only "fall" to the east to get into orbit. In addition, the rotation of the earth will give a small boost to the rocket, as it doesn't need to create this start speed for itself. On average, you will save about 175m/s.

From experience the loss of speed resulting from atmospheric drag will be about 1000m/s. More precise calculations are neither worthwhile nor possible for that matter, as the game only simulates the impact of the surrounding air. For a start in western direction one also should calculate about 300m/s more. For further details, see the youtube video from Mike Aben Ascent Costs of Inclined Orbits - KSP Let's Do The Math.

If you hav the commnet of kerbins surface deactivated, you won't be able to controll the rocket anymore as soon as the KSC loses vision. You can either activate the ground system and keep it on, until your own sattelite commnet stands, or you keep it off and use a kOS-script to release the satellites (more on that later).

If in case you already have a lot of communicating space junk flying around from previous attempts, then you can just use that as a temporary replacement.

1.1 Target Orbit

The target orbit ought to be high enough so that you could draw an imaginary direct line from one satellite to the other without it touching the planet's surface. However, the target orbit should not be too high so not to leave the sphere of influence (SOI) of the planet and sail away into infinity. The best orbit height lies somewhere in between. There is also a special orbit height, in which the satellite needs for one complete rotation around the planet just as long as it takes for the planet to rotate around it's own axis (One Day). Thereby a person standing on the surface of the planet will view the satellite as if it is standing still. The correct term for this kind of orbit is "synchronous orbit".

The KSP Wiki says for the height of a synchronous orbit $2863.334km$. This

means that all objects in this orbit need 5h 59m 9.4s for one complete rotation around the planet.

1.1.1 Alternative Orbit

Tipp: Geostationary, Molniya, Tundra, Polar & Sun Synchronous Orbits Explained - Scott Manley

If you still want a different orbit, then you will have to deal with a little bit of geometry ;)

First we calculate the lowest possible orbit for our chosen number of satellites. The orbit height from the planet's center (radius r) we get through this formular:

$$r_{orbit} = \frac{r_{planet}}{\sin\frac{\theta}{2}} \tag{1.1}$$

The angle θ of the polygon is being calculated by:

$$\theta = \frac{(n-2)\cdot 180°}{n} \tag{1.2}$$

$(n-2)\cdot 180°$ stands for the sum of angles from the regular polygon and n for the number of angles, or rather for the number of satellites in orbit. In our case we have a triangle $\rightarrow$ so $\theta = 60°$ for each of the three interior angles.

$$r_{orbit} = \frac{r_{planet}}{\sin\frac{\theta}{2}} = \frac{600km}{\sin\frac{60°}{2}} = 1200km$$

For us this means that we have got a $1200km - 600km = 600km$ distance from the surface of the planet for the lowest possible orbit that one can achieve with three satellites at the equator.

As time goes by the satellites will slitely change their position between each other, provided that they aren't being maintained or the safefile isn't being manipulated: Kerbal Space Program - Savefile Editing & Orbital Parameters Tutorial (Fixed) - Scott Manley. This means that sooner or later the connection between the satellites will be interrupted by the surface of the planet. For this reason it's advisable to add a small buffer of a few kilometers to a low orbit.

The orbital period of the target orbit (final orbit) is obtained by:

$$T_f = 2\pi\sqrt{\frac{a_f^3}{\mu}} = 2\pi\sqrt{\frac{(1200km)^3}{\mu}} = 4395s \tag{1.3}$$

$\mu = 3.5316\cdot 10^{12}\frac{m^3}{s^2} = 3531.6\frac{km^3}{s^2}$
(kerbin standard gravitational parameter)

In a circular orbit, the semimajor axis a
is equal to the orbit radius r, the periapsis Pe and the apoapsis Ap.

But back to the "Synchronous Orbit".

1.2 Transfer Orbit

(GTO von engl. geosynchronous / geostationary transfer orbit)
Now the question is how actually we want to deploy the satellites into their orbit while at the same time spacing them at equal distances. The youtuber "Stratzenblitz75" has made a great video on this question. KSP/RSS - Building a Comm Network with Math! - [Stratlab EP1]

We measure (or calculate) the time a satellite spends in the target orbit for one complete rotation around the planet. That time is also called "orbital period". In our case we already know from the KSP Wiki that the orbital period is about 6 hours which is the same as one kerbal day. And we want release three equally distanced satellites in this orbit. This means that each satellite has a temporal distance of 2 hours (1/3 of it's orbit) to the next one.

The next thought would be that maybe we could just release the satellites one after the other at the same spot in the target orbit. The released satellite will of course then fly on in his orbit and make place for the next one. If we play our cards right, then we will release the next satellite only after the former one has completed a distance of exactly 1/3 of it's orbit.

This, in turn, is possible by first putting all three satellites onto one single rocket, which will then pass the exact spot periodically. The orbit of the rocket must have the form of an ellipse. The highest part of the orbit is called apoapsis and is exactly the one that passes the target orbit. The lowest part of the orbit is called periapsis (you will also hear things like apogee or perigee) and hase been deliberately chosen to make the orbital period exactly 2/3 of the target orbit. Now, everytime the rocket passes it's apoapsis, a new satellite will be released. After three full circles around the planet, all the satellites will hopefully have been deployed. Now the rocket can begin it's deorbit burn to lower it's periapsis and to burn up in the atmosphere and simultaneously reduce space debris. We do that for safety reasons, but in KSP it primarily improves the runtime performance, as less parts will have to be rendered. Because the population density on kerbin is very low, it isn't a problem to crash your parts into the planet, but in real life you try to make sure that the parts burn up in the atmosphere before hitting the ground. Alternatively the is also a button inside the KSC Observation Center to delete those parts.

If, however, the lowest part of this tranfer orbit goes through the surface of the planet (precisely because the target orbit is set so low), then we can just change it's orbital period to something, that isn't 2/3 but rather 5/6 of the target orbit. This will hopefully place the periapsis above the planet's surface (and it's atmosphere). However it should be noted that in order to reach the goal, now we can only release one satellite every two circles around the planet, instead of every round with a 2/3 orbit. In this case the orbital period ammounts to $5/6 \cdot 6h = 5h$.

If, however, the lower part of this orbit is even now still too close to the planet, then we can decide choose a 4/3 orbit. This way the periapsis will be at the height of the target orbit and the apoapsis higher than that. In this case the orbital period ammounts to $4/3 \cdot 6h = 8h$.

The closer the tranfer orbit is to the target orbit, the better. The satellite will have to spend less energy to get into it's final orbit. This in turn leads to a smaller satellite, as less fuel (and thrust) will be needed.

But where do we now put the transfer orbit? All that is needed are the apoapsis and the periapsis. We already got one of the two. That's the point where the transfer orbit intersects the target orbit. We can now calculate the other point with the help of the orbital period:

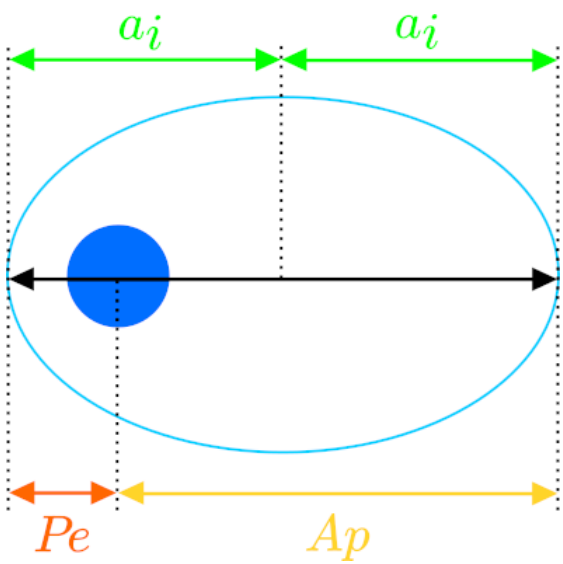


Figure 1.2: Transfer Orbit Parameter

Orbital Period from the Transfer Orbit:

$$
\begin{aligned}
T_i &= \frac{2}{3} T_f \\
&= \frac{2}{3} \cdot 6h \\
&= 14400s
\end{aligned}
\tag{1.4}
$$

Or how the KSP Wiki states it:
$T_f = 5h\,59m\,9.4s = 21549.425s$
$T_i = \frac{2}{3} \cdot 21549.425s = 14366.283s$

Semimajor Axis from the Transfer Orbit:

$$
\begin{aligned}
a_i &= \sqrt[3]{\frac{\mu \cdot T_i^2}{4\pi^2}} \\
&= \sqrt[3]{\frac{3531.6 \frac{km^3}{s^2} \cdot (14366.283s)^2}{4\pi^2}} \\
&= 2643km
\end{aligned}
\tag{1.5}
$$

Periapsis from the Transfer Orbit:

$$
\begin{aligned}
Pe &= 2a_i - Ap \\
&= 2(2643km) - (600km + 2863.334km) \\
&= 1822.7km
\end{aligned}
\tag{1.6}
$$

Here, the orbit height amounts to $1822.7km - 600km = 1222.7km$ above the surface.

Alternatively you can of course just use the online Resonant Orbit Calculator from Eric Meyer. After choosing the planetary body the calculator will present you with the required Δv for each satellite. Especially the visual presentation helps in the understanding.

1.3 Delta-v budget of each satellite

Δv can be understood as the whole change in velocity without the influence of drag and other parameters. Would we simple separate the satellite from it's rocket then it actually might move away a little bit, but the orbit would stay more or less the same. To finally get into the target orbit, the satellite must first burn prograde (in the moving direction) at the apoapsis until it's velocity is as high as are all objects inside of the target orbit. For our "synchronous orbit" this would mean an velocity of $v = \sqrt{\frac{\mu}{r_o}} = 1110m/s$ for every object in this orbit, whereby r_o must be

the radius of a round orbit. The higher the orbit, the higher the potential energy and the lower the speed. Or 2. Kepler's law: "A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.". Objects in a lower orbit must therefore move faster.

How much Δv does the satellite need to get from the tranfer orbit to the target orbit? For this purpose we use the Vis-Viva Equation:

$$\begin{aligned}
\Delta v_2 &= \sqrt{\frac{\mu}{r_2}}\left(1-\sqrt{\frac{2r_1}{r_1+r_2}}\right) \\
&= \sqrt{\frac{3531.6\frac{km^3}{s^2}}{3463.334km}}\left(1-\sqrt{\frac{2(1822.7km)}{1822.7km+3463.334km}}\right) \\
&= 0.171224\frac{km}{s} \\
&= 171.224\frac{m}{s}
\end{aligned}$$

$$r_1 = 1822.7km = Pe$$
$$r_2 = 3463.334km = Ap$$

1.4 Delta-v budget of the rocket

Ascent & Injection Costs - KSP Let's Do The Math #6
Ascent Costs of Inclined Orbits - KSP Let's Do The Math #7

The rocket will execute three burn maneuvers:

1. Start from the surface of kerbin, until the apoapsis is 1822.7km (1222.7km above ground).
2. Orbit circularisation burn at the apoapsis to raise the periapsis to the same height.
3. Raising the apoapsis to the height of the target orbit at 3463.334km (2863.334km above ground).

Of course one can also achieve the same with only two maneuvers, but then the accuracy would suffer. In addition one would waste a lot of fuel, because the longer the burn, the less the time at which the rocket will actually burn into the prograde direction.

Sidereal rotational velocity of kerbin: $\overrightarrow{v_r} = 174.94m/s$
Air drag by experience: $\overleftarrow{drag} = 1000m/s$
Radius of Kerbin: $r_1 = 600km$
Radius from the first preliminary orbit: $r_2 = 1822.7km$
Radius from the target orbit: $r_3 = 3463.334km$

Let's now calculate the necessary energy for the first level:

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1+r_2}} - 1\right) = 549.89m/s$$

$$\Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1+r_2}}\right) = 412.3m/s$$

$$\Delta v_1 = \sqrt{\frac{\mu}{r_2}}\left(\sqrt{\frac{2r_3}{r_2+r_3}} - 1\right) = 201.4m/s$$

$$\begin{array}{r} 549.89 \\ +412.3 \\ +201.4 \\ \hline 1163.59 \end{array}$$

Until now we pretended that the rocket will start from an orbit with the height of 600km. But of course the rocket starts actually from the ground's surface with the speed of the planet's own rotation. Thus, we still need the necessary velocity for an orbit of 600km, plus the speed which was lost due to air drag inside the atmosphere, minus the additional rotation speed of the planet at a launch to the east.

Theoretical orbital velocity at ground level:

$$\Delta v = \sqrt{\frac{\mu}{600km}} = 2426.1m/s$$

$$\begin{array}{r} 2426.1 \\ +1000 \\ -\ 174.94 \\ \hline 3251.16 \end{array}$$

Therefore the rocket requires a minimum Δv of

$$1163.59m/s + 3251.16m/s = \mathbf{4414.75m/s}.$$

If you really want to fly as economically as possible, then you should start with the very lowest possible orbit above the atmosphere, somewhere in between 70-80km. From there on you should raise the apoapsis "directly" to the height of the target orbit. After arriving at the apoapsis, the rocket should then raise it's periapsis for the correct transfer orbit.

In this case you optimally want to raise the orbit only bit by bit by using the so-called "orbit phasing" method, just as Rocket Lab does it with it's Photon Kick Stage on it's way to the moon.

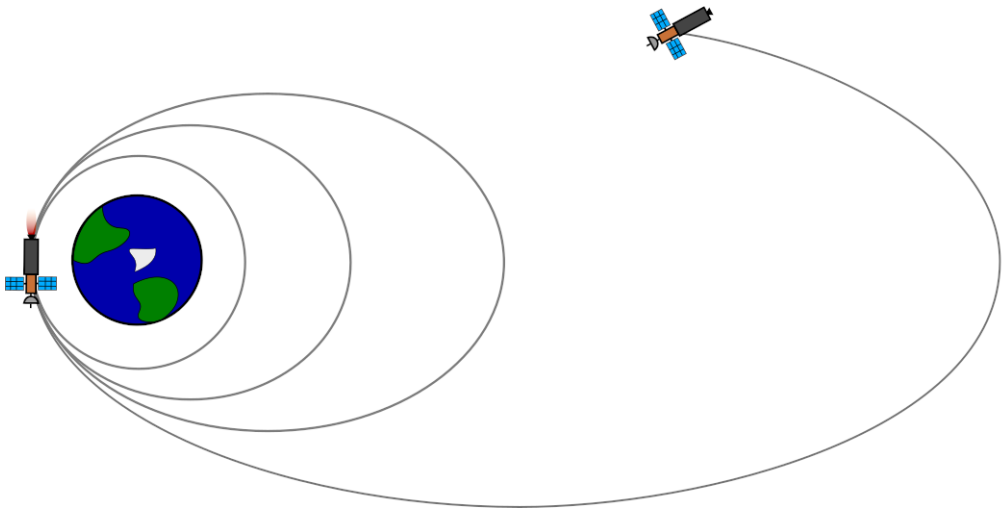


Figure 1.3: Orbit Phasing Maneuver

2 Mun Satellites

Mun is the closest neighbor to Kerbin. Strong antennas are therefore less relevant than a continuous communication connection with the vessels that might lie inside the mun shadow as seen from kerbin (behind Mun). At worst case scenario, a loss of connection will lead to a complete loss of control. This is particularly relevant if you want to land on the moon's surface.

Because the approach does not differ a lot from the one used on the Kerbin satellites, I won't go into detail as much.

Mun Properties:
Equatorial radius $= 200km$
$\mu = 6.5138398 \cdot 10^{10} \frac{m^3}{s^2} = 65.1384 \frac{km^3}{s^2}$
SOI (Sphere of influence) $= 2429.5591km$

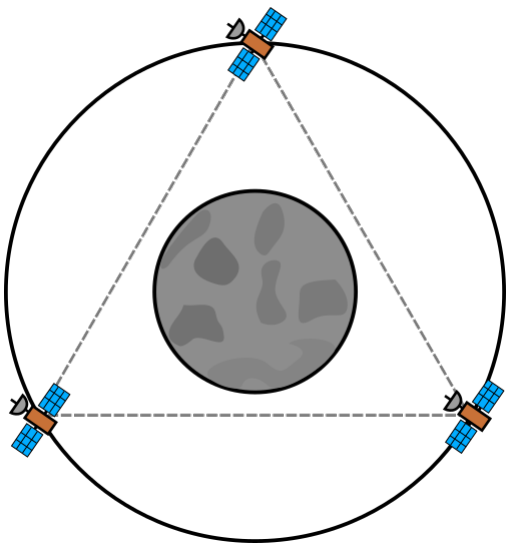


Figure 2.1: Mun Relay Network

Minimal Orbit:

$$r_{orbit} = \frac{r_{mun}}{\sin\frac{\theta}{2}} = \frac{200km}{\sin\frac{60°}{2}} = 400km$$

$400km - 200km = 200km$ is the distance from the moon's surface for the lowest possible orbit with three satellites around the equator.

Maximum Orbit:

The maximum orbit lies shortly below the SOI, in other words, below the distance from which on Kerbin's pull starts to outweigh that of the moon. Also, from exceeding this "threshold" on, the point of reference is being switched for all the following calculations.

Synchronous Orbit:

Unfortunately, a synchronous orbit is not possible as the orbit radius lies beyond Mun's SOI.

Optimum Orbit:

Mike Aben specifies for the ideal orbit height a value of $337.35km$ and for the periapsis a value of $103.85km$. Of course the apoapsis has the same height as the target orbit.
The reason: Mike Aben wants the satellites to be as small and with as few antennas as possible while maintaining a communication strength of about 80%. I can only recommend The Best Relay Orbit - KSP Let's Do The Math #13 just like the other videos on this channel.

2.1 Delta-v budget of each satellite

How much Δv does the satellite need to get from the tranfer orbit to the target orbit? For this purpose we use the Vis-Viva Equation:

$$r_1 = 103.85km + 200km = 303.85km = Pe$$
$$r_2 = 337.35km + 200km = 537.35km = Ap$$

$$\begin{aligned}
\Delta v_2 &= \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}}\right) \\
&= \sqrt{\frac{65.1384\frac{km^3}{s^2}}{537.35km}}\left(1 - \sqrt{\frac{2(303.85km)}{303.85km + 537.35km}}\right) \\
&= 0.0522416\frac{km}{s} \\
&= 52.2416\frac{m}{s}
\end{aligned}$$

2.2 Delta-v budget of the rocket

We already know from the previous chapter that we require $3251.16m/s$ for a start from the surface of Kerbin. This figure was arrived at because of us launching from the ground instead of an orbit. Therefore we need to add the required velocity that is needed for an imaginary orbit with a radius of 600km (ground level), plus the speed which was lost due to air drag inside the atmosphere, minus the additional rotation speed of the planet at a launch to the east.

The rocket will execute four burn maneuvers:

1. Start from the surface of Kerbin, until the apoapsis is $685km$ ($85km$ above ground).

$$\Delta v_1 = \sqrt{\frac{3531.6\frac{km^3}{s^2}}{600km}}\left(\sqrt{\frac{2(685km)}{600km+685km}}-1\right) = 78.956m/s$$

2. Orbit circularisation burn at the apoapsis to raise the periapsis to the same height.

$$\Delta v_2 = \sqrt{\frac{3531.6\frac{km^3}{s^2}}{685km}}\left(1-\sqrt{\frac{2(600km)}{600km+685km}}\right) = 76.382m/s$$

3. Raising the apoapsis to the height of Mun ($12000km$).

$$\Delta v_1 = \sqrt{\frac{3531.6\frac{km^3}{s^2}}{685km}}\left(\sqrt{\frac{2(12000km)}{685km+12000km}}-1\right) = 852.61m/s$$

The "orbital inclination" is 0°. This means that Mun's orbit lies exactly above Kerbin's equator.

Velocity at apoapsis with Kerbin as point of reference:

$$\begin{aligned} v &= \sqrt{\mu\left(\frac{2}{r}-\frac{1}{a}\right)} \\ &= \sqrt{3531.6\frac{km^3}{s^2}\left(\frac{2}{12000km}-\frac{1}{6342.5km}\right)} \\ &= 178.283m/s \end{aligned}$$

4. Mun capture burn to fall into an orbit around Mun.

   The speed, at which Mun circles around Kerbin, is about $543m/s$. Minus the speed that the rocket already has at the height of the orbit of Mun, we get a velocity difference of $543m/s - 178.283m/s = 364.717m/s$. From entering the SOI of the moon on the rocket will begin to be pulled in by the moon's gravity. It is also relevant from which direction the rocket is coming and also the altitude at which it is passing above the surface. It is advisable to always include about 5-10% more Δv than necessary (especially for landing attempts on the surface of Mun), but he who is interested might also want to become familiar with the so-called Oberth Effect. This is also directly related to the topic so so-called Gravity Assists which is especially useful for interplanetary missions.
   Alternatively you might want to take a look at the Kerbin Delta-v Map from the KPS-Community. On it, the Δv values are listet in the same way like one a train timetable. For all those mathematicians between you, who want to know the exact origin of these values, Mike Aben has got you covered again: Calculating Transfer & Capture Costs - The Mun | KSP Let's Do The Math

   The steps are:

   a) Δv_2 (6.2) for calculating the velocity differential compared to Mun on exactly the right altitude above Kerbin in order to make rocket and moon intersect each other. Optionally look above.

$$
\begin{aligned}
\Delta v_2 &= \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}}\right) \\
&= \sqrt{\frac{3531.6\frac{km^3}{s^2}}{12000km}}\left(1 - \sqrt{\frac{2(685km)}{685km + 12000km}}\right) \\
&= 0.36421km/s \\
&= 364.21m/s
\end{aligned}
$$

b) Calculate rocket velocity at the lowest flyby above the moon:

$$
\begin{aligned}
v_1 &= \sqrt{2\mu\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + v_2^2} \\
&= \sqrt{2(65.1384\frac{km^3}{s^2})\left(\frac{1}{303.85km} - \frac{1}{2429.559km}\right) + (0.365km/s)^2} \\
&= 0.71299km/s \\
&= 713m/s
\end{aligned}
\tag{2.1}
$$

c) Calculate rocket velocity at the same spot, but within the transfer orbit (rocket's final orbit):

$$
\begin{aligned}
v_2 &= \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)} \\
&= \sqrt{65.1384\frac{km^3}{s^2}\left(\frac{2}{303.85km} - \frac{1}{420.6km}\right)} \\
&= 0.5233387774km/s \\
&= 523.34m/s
\end{aligned}
$$

d) Difference of flyby-velocity and transfer orbit velocity at the same spot in space:

$$713m/s - 523.34m/s = 189.66m/s$$

Therefore the rocket requires $\mathbf{\Delta v \approx 190m/s}$ *in order to insert into a munar orbit with a periapsis of* $303.85km$ *and an apoapsis of* $537.35km$*. The Delta-v Map lists a value of 310m/s, but that's because the 214km orbit that was used there is very low.*

$$
\begin{array}{r}
3251.16 \\
+\quad 78.96 \\
+\quad 76.38 \\
+\quad 852.61 \\
+\quad 190 \\
\hline
4449.11
\end{array}
$$

Therefore the rocket must achieve a velocity difference of around **4450m/s**.

3 Minmus Satelliten

Minmus Properties:
Equatorial radius = $60km$
$\mu = 1.7658000 \cdot 10^9 \frac{m^3}{s^2} = 1.7658 \frac{km^3}{s^2}$
SOI (Sphere of influence) = $2247.4284km$

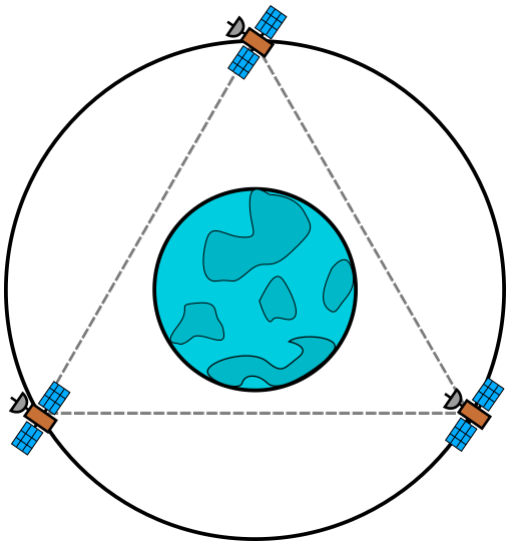


Figure 3.1: Minmus Relay Network

Because with $47000km$ distance to it's home planet, Minmus is quite a bit further away than even Mun, we will already be thinking about using stronger antennas than would be required for only the communication between the three satellites.

Minimum Orbit:

The distance from the surface of Minmus for the lowest possible orbit with three satellites at the equator is $60km$.

Synchronous Orbit:

With $357.941km$ it is located far below the SOI.

Optimum Orbit:

The distance of the synchronous orbit creates an exceptionally large viewing angle onto the moon's surface. Furthermore the very low velocity at this altitude very much simplifies releasing the satellites accurately into their final orbit. The tranfer orbit has got an apoapsis of $357.941km$ and a periapsis of $159.9564km$. Alternatively you can use the online Resonant Orbit Calculator from Eric Meyer for assistance.

3.1 Delta-v budget of each satellite

$$r_1 = 159.9564km = Pe$$
$$r_2 = 357.941km = Ap$$

$$\begin{aligned}
\Delta v_2 &= \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}}\right) \\
&= \sqrt{\frac{1.7658\frac{km^3}{s^2}}{357.941km}}\left(1 - \sqrt{\frac{2(159.9564km)}{159.9564km + 357.941km}}\right) \\
&= 0.0150343km/s \\
&= 15m/s
\end{aligned}$$

3.2 Delta-v budget of the rocket

The rocket will execute four burn maneuvers:

1. Start from the surface of Kerbin, until the apoapsis is $685km$ ($85km$ above ground).

   $\Delta v_1 = 78.956m/s$ (orbit change) including $3251.16m/s$ (drag and planet rotation) $= 3330.116m/s$

2. Orbit circularisation burn at the apoapsis to raise the periapsis to the same heigth.

   $\Delta v_2 = 76.382m/s$

3. Raising the apoapsis to the height of Minmus ($47000km$).

$$\Delta v_1 = \sqrt{\frac{3531.6\frac{km^3}{s^2}}{685km}}\left(\sqrt{\frac{2(47000km)}{685km+47000km}}-1\right) = 917.36m/s$$

The "orbital inclination" of the orbit of Minmus in comparison to Kerbin's equator is 6°. If you go about it the right way then you want to start directly into an orbit that is tilted to about 6°. If that's too difficult (because one also has to consider the perfect timing) you can first launch to the east and after reaching space turn the orbit plane to about 6°, so that when viewed from the side the orbit of the rocket and the orbit of the moon together make a single line. However, turning the orbit plane is very costly so that you will have to budget about $340m/s$ more as one can see on the Delta-v Map.

4. Minmus capture burn to fall into an orbit around Minmus.

   a) Δv_2 (6.2) for calculating the velocity differential compared to Minmus on exactly the right altitude above Kerbin in order to make rocket an moon intersect each other.

$$\begin{aligned}\Delta v_2 &= \sqrt{\frac{\mu}{r_2}}\left(1-\sqrt{\frac{2r_1}{r_1+r_2}}\right)\\ &= \sqrt{\frac{3531.6\frac{km^3}{s^2}}{47000km}}\left(1-\sqrt{\frac{2(685km)}{685km+47000km}}\right)\\ &= 0.227655km/s\\ &= 227.66m/s\end{aligned}$$

   b) Calculate rocket velocity at the lowest flyby above the moon:

$$\begin{aligned}v_1 &= \sqrt{2\mu\left(\frac{1}{r_1}-\frac{1}{r_2}\right)+v_2^2}\\ &= \sqrt{2(1.7658\frac{km^3}{s^2})\left(\frac{1}{159.9564km}-\frac{1}{2247.4284km}\right)+(0.22766km/s)^2}\\ &= 0.26895km/s\\ &= 268.95m/s\end{aligned} \tag{3.1}$$

c) Calculate rocket velocityy at the same spot, but within the transfer orbit (rocket's final orbit):

$$
\begin{aligned}
v_2 &= \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)} \\
&= \sqrt{1.7658\frac{km^3}{s^2}\left(\frac{2}{159.9564km} - \frac{1}{258.9487km}\right)} \\
&= 0.12353km/s \\
&= 123.53m/s
\end{aligned}
$$

d) Difference of flyby-velocity and transfer orbit velocity at the same spot in space:

$$268.95m/s - 123.53m/s = 145.42m/s$$

Therefore the rocket requires $\mathbf{\Delta v \approx 145m/s}$ *in order to insert into a Minmus orbit with a periapsis of* $159.9564km$ *and an apoapsis of* $357.941km$.

$$
\begin{array}{r}
3330.12 \\
+\quad 76.38 \\
+\quad 917.36 \\
+\quad 145 \\
\hline
4468.86
\end{array}
$$

Therefore the rocket must achieve a velocity difference of around **4470m/s**. With an additional orbit plane change of 6° above Kerbin, this adds about $340m/s$ and thus results in **4810m/s**.

4 Building rockets and satellites

Kerbin

Rocket ($\Delta v = 4415 m/s$) with 3 satellites (each $\Delta v = 172 m/s$)

Mun

Rocket ($\Delta v = 4450 m/s$) with 3 satellites (each $\Delta v = 53 m/s$)

Minmus

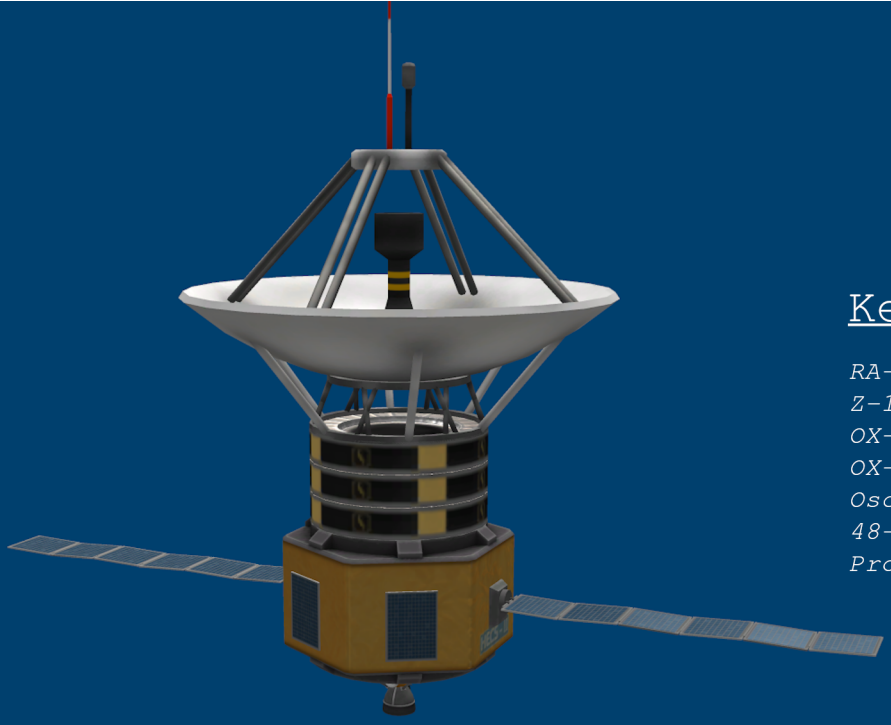
Rocket ($\Delta v = 4470 m/s$) with 3 satellites (each $\Delta v = 15 m/s$)

Electricity supply for each satellite

- Calculating Dark Side Time
- Determining Electricity Needs

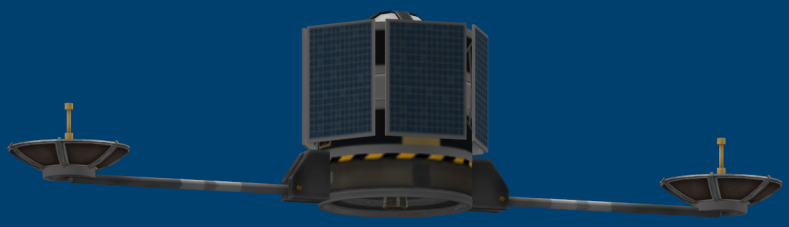
Antenna signal strength for each satellite

Communication between the three satellites and with the KSC. CommNet Wiki
A very helpful Excel document for for calculating the necessary anntenna signal strength you find at CommNet Signal Strength Calculator & Antenna Selector, or https://goo.gl/Wn03VL



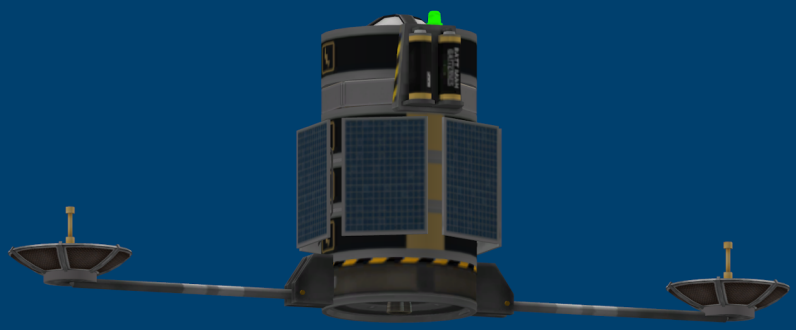
Kerbin Sat

RA-100 Relay Antenna (1)
Z-1k Rechargeable Battery Bank (3)
OX-4L 1x6 Photovoltaic Panels (2)
OX-STAT Photovoltaic Panels (4)
Oscar-B Fuel Tank (1)
48-7S "Spark" Liquid Fuel Engine (1)
Probodobodyne HECS2 (1)



Mun Sat

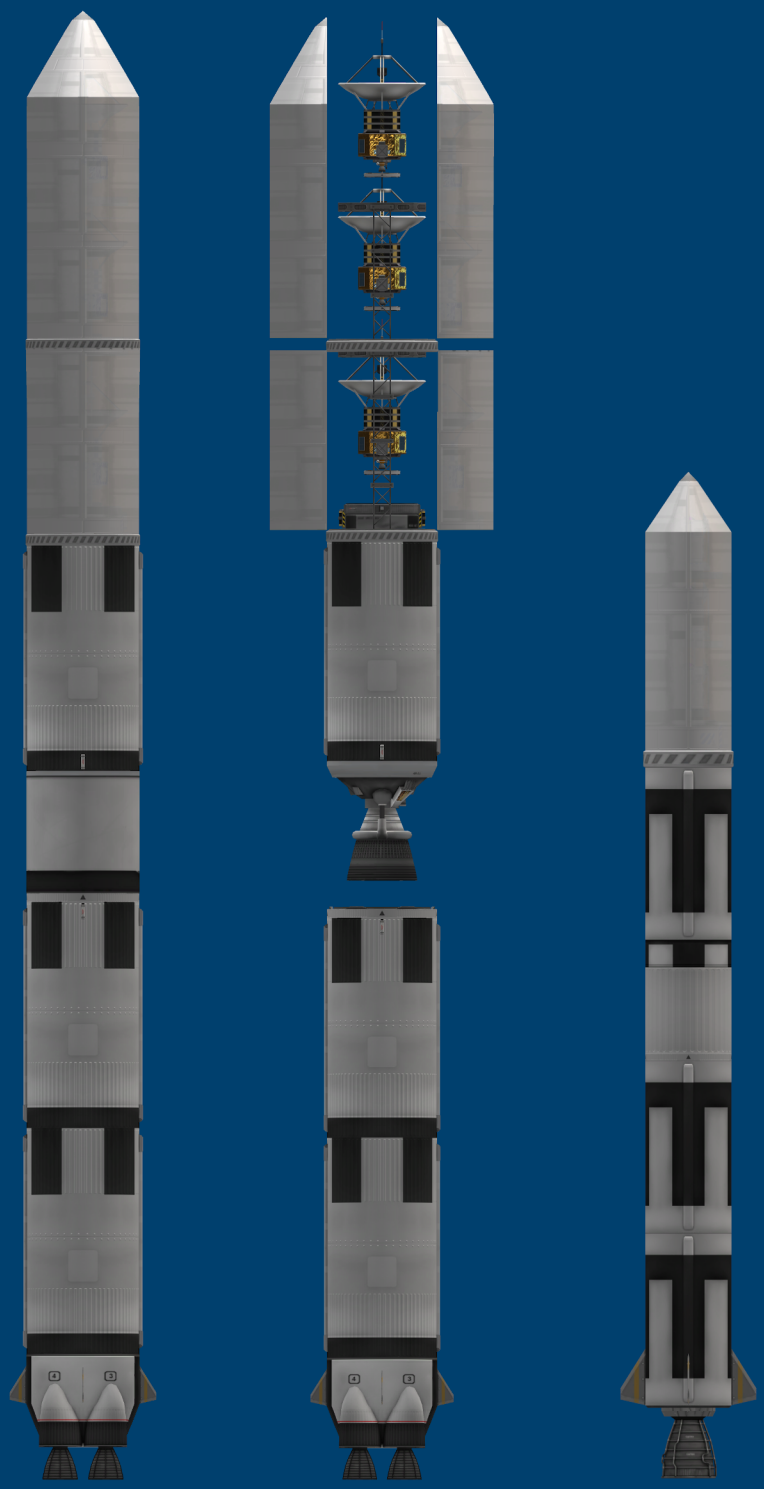
HG-5 High Gain Antenna (2)
Z-200 Rechargeable Battery Bank (2)
OX-STAT Photovoltaic Panels (6)
Stratus-V Round. Monoprop. Tank (1)
Place-Anywhere 7 Linear RCS Port (1)
Probodobodyne OKTO2 (1)
Small Inline Reaction Wheel (1)



Minmus Sat

HG-5 High Gain Antenna (2)
Z-200 Rechargeable Battery Pack (4)
Z-100 Rechargeable Battery Pack (2)
OX-STAT Photovoltaic Panels (6)
Stratus-V Round. Monoprop. Tank (1)
Place-Anywhere 7 Linear RCS Port (1)
Probodobodyne OKTO2 (1)
Small Inline Reaction Wheel (1)

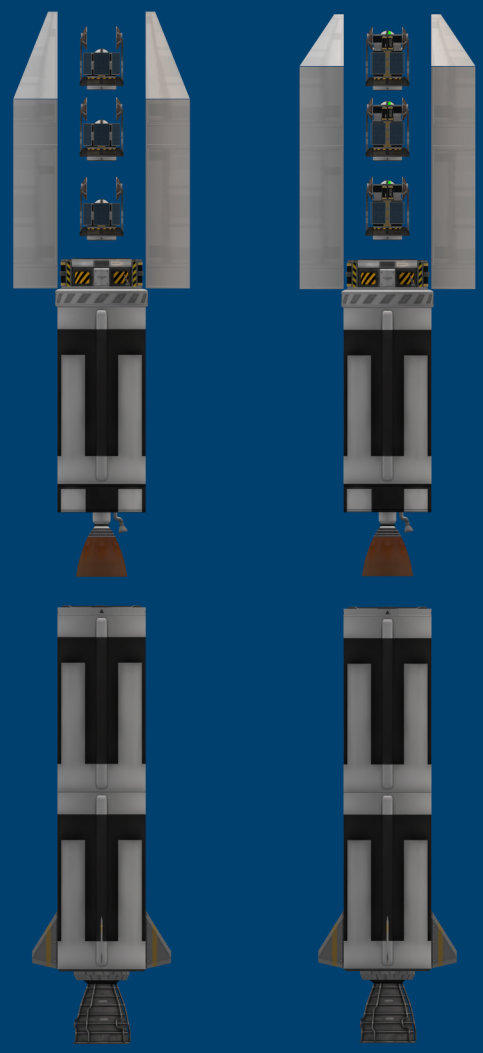
Kerbin Booster



Second Stage
- Kerbodyne KR-2L+ "Rhino"
  Liquid Fuel Engine
- Kerbodyne S3-14400 Tank
- Advanced Reaction Wheel Module, Large
- RC-L01 Remote Guidance Unit
- Z-400 Rechargeable Battery

First Stage
- S3 KS-25x4 "Mammoth"
  Liquid Fuel Engine
- Kerbodyne S3-14400 Tank

Moon Boosters



Second Stage
- LV-T91 "Cheetah"
  Liquid Fuel Engine
- FL-TX1800 Fuel Tank
- Advanced Inline Stabilizer
- RC-001S Remote Guidance Unit
- Z-400 Rechargeable Battery

First Stage
- S3 KS-25 "Vector"
  liquid fuel engine
- FL-TX1800 Fuel Tank

Helper Tools

Launch Window Planner:

- https://ksp.olex.biz/
- https://alexmoon.github.io/ksp/

Delta-V Planner:

- https://13375.de/KSPDeltaVMap/

Resonant Orbit Calculator:

- https://meyerweb.com/eric/ksp/resonant-orbits/

KSP Wiki list:

- https://wiki.kerbalspaceprogram.com/wiki/Calculation_tools

Addons:

- https://spacedock.info/kerbal-space-program

5 Programming the rocket

5.1 Kerbal Operating System (kOS)

kOS is a community project of the Kerbal Space Program fanbase. It's basically an autopilot that you have to write for yourself. The mod introduces a few own parts into the game, of which at least one has to be put onto the vessel in order to work. Each part consists of a "scriptable control system" and a limited storage medium. Also keep in mind that the control chip will require additional power.

The File-Endung for Kerboscript is always "ks", for example "launch.ks". When opening the kOS-console in the simulator, one can then switch to the "harddrive" of the spacecraft by typing

```
SWITCH TO 1.
```

1 stands for the vessel and *0* for the Kerbal Space Center. Because it seems unrealistic for the rocket to be always control by the KSC, most players will transfer their script to the rocket before launching.

```
list.
```

shows the scripts that lie on the active harddrive. Keep in mind that the period at the end of an instruction is part of the syntax. Without it you will only receive error messages.

```
COPYPATH("0:launch", "").
```

copies the script "launch.ks" from the harddrive *0* (KSC) onto the active harddrive (here *1*).

```
RUN launch. // or else
RUNPATH("launch").
```

both start the script. The "runpath" method is simply a newer version. The following lines of code allow for the use of additional parameters set afterwards by the user.

```
runpath("launch",90,90000,0.35). // or else
run launch(90,90000,0.35).
```

The parameter **90** stands for the compass direction of rotation "east", **90000** for the target orbit height in metern and **0.35** for the tilting speed at the beginning, because the script is supposed to be used for different rocket types, which requiers the ability to use slightly different settings.

For the beginning I want to suggest the tutorial-series from CheersKevin - kOS for Newbies. Even suitable for future programmers! Not only can you launch rockets with those scripts, but also drones and even Airplanes. Two very impressive examples see here: HoverBot drone, SpaceX Falcon Heavy.

5.2 Reaching LEO - Low Earth Orbit

Reaching the first stable orbit is always a challenge, here as well in the real world. Apart from the gravitational pull, the rocket also has to fight aero dynamics and if you reach a too high velocity in too low of an altitude, even with heat and g-force. Furthermore the rocket has to be balanced by positioning the "center of lift" below the "center of mass". So just like a dart. Of course it helps a lot that most engines have the ability to "gimble" - that is, they can rotate the nozzle into multiple directions. In this way they can balance the rocket without the need of wings. In the real world you also want to pay attention to the weather in high altutudes. That's because in some heights, winds can reach high speeds of about 1000km/h, which is enough to split a rocket in half. Additionally there are also risks of lightning strikes because in different charged air layers the rocket might act just like a lightning rod. This in turn could potentially damage onboard circuits. Apollo 12 was victim to two such lightning strikes but was lucky insofar as it didn't create any permanent damage.

In you ever watched a rocket launch, then you might have noticed that the rocket didn't just fly straight up, but that it began to tilt sideways after just a few seconds. This is called the "pitching maneuver".
But why are rockets doing that?

The only reason why an object can stay in orbit is because it's parallel velocity to the surface is greater than the attraction in the direction of the planet. In this case the object will miss the earth while being pulled towards it, because it's speed is so fast that at the point, at which the planet has pulled the object 50 meters

downwards, the curvature of the earth will also have “made the ground 50 meters lower”. Back to the rocket..
At low altitudes the rocket is constantly being decelerated by the atmosphere. Therefore one would normally assume that it would be the best option to leave the atmosphere as early as possible by flying straight up until it’s in space. Following that, the rocket now only needs to generate speed parallel to the surface. The problem is that it normally takes a lot of time to reach the required speed of about 8km/s. In other words, you have to fly really high to have enough time left in order to reach orbital velocity. You might already have noticed that this method wastes quite a lot of energy as the horizontal speed isn’t what makes one stay in orbit. That’s why people decided to use a curved trajectory which begins vertically and becomes horizontal over time.
Again: The rocket only goes “up” to get out of the atmosphere. If the rocket was standing on the surface of the moon, then it would only have to accelerate sideways. The only constraint might be the height of the mountains or the power of the engine (thrust to weight ratio)

5.2.1 kOS Gravity Turn

```
parameter compass is 90, finalApoapsis is 85000, pitchFactor is
    0.4.

function main {

  CLEARSCREEN.
  print "compass heading is " + compass + "degrees".
  print "finalApoapsis is " + finalApoapsis + "m".
  print "pitchFactor is " + pitchFactor.

  doLaunch().
  doAscent().
  until apoapsis > finalApoapsis {
    doAutoStage().
  }
  doShutdown().
  doCirculate().

  print "program exited.".
}

//--------------
//   LAUNCH:
//--------------

```

```
function doSafeStage {
  wait until stage:ready.
  stage.
  print "staging.".
}

function doLaunch {
  PRINT "Counting down:".
  FROM {local countdown is 5.} UNTIL countdown = 0 STEP {SET countdown to countdown - 1.} DO {
    PRINT "..." + countdown.
    WAIT 1.
  }
  lock throttle to 1.
  doSafeStage().
  lock steering to heading(90, 90, 270).
}

function doAscent {
  set targetPitch to 90.
  set targetDirection to compass.
  set targetRoll to 0.
  wait until verticalSpeed >= 60.
  print "pitching maneuver started.".
  until targetPitch < (88.963 - 1.03287 * alt:radar^0.409511) {
    set targetPitch to targetPitch - pitchFactor.
    lock steering to heading(targetDirection, targetPitch, targetRoll).
    wait 0.5.
  }
  lock targetPitch to 88.963 - 1.03287 * alt:radar^0.409511.
  lock steering to heading(targetDirection, targetPitch, targetRoll).
}

function doAutoStage {
  if not(defined oldThrust) {
    declare global oldThrust to ship:availablethrust.
  }
  if ship:availableThrust < (oldThrust - 10) {
    doSafeStage(). wait 1.
    set oldThrust to ship:availablethrust.
  }
}

function doShutdown {
  lock throttle to 0.
  // lock steering to prograde + R(0,0,270).
  lock steering to prograde.
```

```
  print "shutting down and holding prograde.".
}

//--------------
//  CIRCULATE:
//--------------

function doCirculate {
  until ship:altitude > 70005 {
    PRINT "SHIP:APOAPSIS " + ROUND(SHIP:APOAPSIS,0) AT (0,18).
    PRINT "SHIP:PERIAPSIS " + ROUND(SHIP:PERIAPSIS,0) AT (0,19).
  }
  // wait until ship:altitude > 70005.
  print "Calc. Circulation Maneuv.".
  set burnTime to getManeuverBurnTime().
  // Set the start and end times.
  set start_time to abs(time:seconds + eta:apoapsis - (burnTime / 2)).
  set end_time to abs(time:seconds + eta:apoapsis + (burnTime / 2)).
  WAIT UNTIL TIME:SECONDS >= start_time.
  LOCK throttle TO 1.
  print "burn started.".
  UNTIL TIME:SECONDS >= end_time {
    // doAutoStage() function from above with small changes
    if not(defined oldThrust) {
      declare global oldThrust to ship:availablethrust.
    }
    if ship:availableThrust < (oldThrust - 10) {
      set end_time to end_time + 2. // added because of lost time while staging
      doSafeStage(). wait 1.
      set oldThrust to ship:availablethrust.
    }
  }
  LOCK throttle TO 0.
  print "burn finished.".
  unlock all.
  SET SHIP:CONTROL:PILOTMAINTHROTTLE TO 0.
}

function getManeuverBurnTime {
  // local my is constant:G * Kerbin:Mass. // Alternative
  local my is 3531.6.
  local r is 600 + (finalApoapsis / 1000).
  local a is (ship:apoapsis + ship:periapsis + 1200000) / 2000.
  // The Velocity of any object in that orbit.
  local Vf is sqrt(my / r).
  local Vi is sqrt(-(my / a) + ((2 * my) / r)).
```

```
  // Velocity-Difference from one orbit to the other.
  local dV is (Vf - Vi) * 1000.

  print "calculated dV: " + CEILING(dV,2) + "m/s.".
  local g0 is 9.80655.
  local isp is 0.

  list engines in myEngines.
  for en in myEngines {
    if en:IGNITION and not en:FLAMEOUT {
      set isp to isp + (en:isp * (en:availablethrust / ship:
    availablethrust)).
    }
  }

  local mf is ship:mass / constant():e^(dV / (isp * g0)).
  local fuelFlow is ship:availablethrust / (isp * g0).
  local t is abs((ship:mass - mf) / fuelFlow).

  print "maneuverBurnTime: " + CEILING(t,2) + "s.".
  return t.
}

main().
```

Listing 5.1: launch.ks

The script shown here is also available on GitHub. The beginning is sometimes difficult. Nevertheless, I want to encourage you to try to write your own script. It's not that difficult. Also there are already many good examples which might help you start....for example this page.
Alternatively there is the mod Smart Parts from linuxgurugamer, for which you don't even need to know how to program. I myself often use this mod in combination, where kerboscript handles the launch and circulation burn and the smart parts stage the fairing and other things that might differ from one rocket to the other.
The following script just executes the next maneuver node on the active vessel's orbit. A big thanks to CheersKevin.

5.3 Next maneuver

```
//--------------
//  MANEUVER:
//--------------
set mnvTime to 0.
set throttleTime to 0.
```

```
function executeManeuverNode {
  local mnv is nextNode.
  set startTime to calculateStartTime(mnv).
  set throttleTime to startTime + mnvTime - 0.5.
  wait until time:seconds > startTime - 30.
  lockSteeringAtManeuverTarget(mnv).
  wait until time:seconds > startTime.
  lock throttle to 1.
  wait until isManeuverComplete(mnv).
  lock throttle to 0.
  removeManeuverFromFlightPlan(mnv).
}

function calculateStartTime {
  parameter mnv.
  print "Node in: " + round(mnv:eta) + "s, DeltaV: " + round(mnv:deltav:mag) + "m/s.".
  set mnvTime to maneuverBurnTime(mnv).
  return time:seconds + mnv:eta - mnvTime / 2.
}

function maneuverBurnTime {
  parameter mnv.
  local dV is mnv:deltaV:mag. //Or by using VisVivaEquation
  local g0 is 9.80665.
  local isp is 0.

  list engines in myEngines.
  for en in myEngines {
    if en:IGNITION and not en:FLAMEOUT {
      set isp to isp + (en:isp * (en:availablethrust / ship:availablethrust)).
    }
  }

  local mf is ship:mass / constant():e^(dV / (isp * g0)).
  local fuelFlow is ship:availablethrust / (isp * g0).
  local t is (ship:mass - mf) / fuelFlow.

  print "maneuverBurnTime: " + round(t) + "s.".
  return t.
}

function lockSteeringAtManeuverTarget {
  parameter mnv.
  lock steering to mnv:burnvector.
}
```

```
function isManeuverComplete {
  parameter mnv.
  if time:seconds > throttleTime {
    lock throttle to 0.2.
  }
  if not(defined originalVector) or originalVector = -1 {
    declare global originalVector to mnv:burnvector.
  }
  if vAng(originalVector, mnv:burnvector) > 90 {
    declare global originalVector to -1.
    return true.
  }
  return false.
}

function removeManeuverFromFlightPlan {
  parameter mnv.
  remove mnv. // 'add mnv.' to add maneuver node
}

executeManeuverNode().
```

Listing 5.2: mnv.ks

For decoupling the satellites from their rocket one could use a timer that uses the orbital period as basis. The satellite could then activate it's engine when the apoapsis or the start time has been reached.

---

6 Orbital Mechanics Crash Course

6.1 Vis-Viva Equation

Calculating the Δv budget for a Hohmann Transfer (Hohmann Transfer).

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1+r_2}} - 1\right) \tag{6.1}$$

$$\Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1+r_2}}\right) \tag{6.2}$$

Δv_1 stands for the cost of the first burn that raises one side of the orbit to the height of the target orbit.

Δv_2 stands for the cost of the second burn that will be executed at the intersection with the target orbit and will transfer the current orbit towards the target orbit.

Together they give the the Δv budget of each spacecraft that transfers from one circular orbit into another circular orbit.
Instead of $\sqrt{\frac{2r}{r_1+r_2}}$ one can also write $\sqrt{\frac{r}{a}}$. The original fraction divided by two shows the radius as the numerator and the semimajor axis as the denominator.
For learning the origin of this formular I suggest the video from Mike Aben. Or perhaps on Wikiwand: Hohmann transfer orbit.

Die Vis-Viva-Gleichung für die Momentangeschwindigkeit eines Körpers, der sich auf einer Bahn um einen anderen Körper befindet, lautet:

$$v = \sqrt{GM\left(\frac{2}{r} - \frac{1}{a}\right)} \tag{6.3}$$

Dabei ist r der Abstand der beiden Körper, v die Relativgeschwindigkeit zwischen den Körpern, a die große Halbachse der Bahn ($a > 0$ für eine Ellipse, $a \to \infty$ für eine Parabel und $a < 0$ für eine Hyperbel), G die Gravitationskonstante und m sowie M die Massen der beiden Körper.

In „Orbital Mechanics“ ist der „standard gravitational parameter“ eines „Astronomischen Objekts“ das Produkt der Gravitationskonstante G und der Masse M des Körpers. Meist wird der griechische Buchstabe μ an Stelle von GM verwendet.
Interessant ist, dass in Orbital Mechanics vorwiegend die Geschwindigkeit zur Berechnung verwendet wird und dabei die Masse eines Raumgefährts eher in den Hintergrund tritt, was sich natürlich durch die enormen Massen der Planeten erklären lässt.
Ist $a = r$ für alle r, entartet die Kepler-Bahn zu einer Kreisbahn; der Körper besitzt überall den gleichen Abstand r vom Schwerpunkt und entsprechend überall dieselbe Geschwindigkeit

$$v_1 = \sqrt{\frac{G(m+M)}{r}} = \sqrt{\frac{\mu}{r}} \tag{6.4}$$

die *Kreisbahngeschwindigkeit* oder erste kosmische Geschwindigkeit.
Damit ein Körper den Einfluss des Zentralgestirns überwinden kann, muss die große Halbachse unendlich groß werden, es gilt also mit $a \to \infty$:

$$v_2 = \sqrt{\frac{2G(m+M)}{r}} = v_1 \cdot \sqrt{2} \tag{6.5}$$

Diese Geschwindigkeit heißt Fluchtgeschwindigkeit oder zweite kosmische Geschwindigkeit. Die Umlaufbahn des Körpers ist dann nicht mehr geschlossen, sondern offen. Ist die Umlaufgeschwindigkeit des Körpers dabei genau gleich v_2, ist die Umlaufbahn eine Parabel, bei größeren Umlaufgeschwindigkeiten dagegen (bei ansonsten gleichbleibendem Abstand r eine Hyperbel und die große Halbachse a wird (formell) negativ.

6.2 Anderer Kram

Space-Hardware, das heißt Raumschiffe, Satelliten und sondergleichen, bleibt nicht etwa deshalb im Weltall, weil es dort draußen keine Gravitation gäbe und man sich im Weltall automatisch in Schwerelosigkeit befände. Tatsächlich ist die Anziehungskraft der Erde in der Entfernung des Mondes nicht sonderlich viel geringer, als auf der Erdoberfläche. Würde man beispielsweise einen Weltraumfahrstuhl bauen, der bis über die dichte Atmosphäre ins „Weltall“ hinausreicht, hätte man auf der obersten Plattform immernoch den selben Erdanziehungseindruck...vielleicht 0.01% weniger oder so. Sollte jemand dort von der Brüstung stolpern, dann würde er schnurgerade nach unten fallen. Dennoch könnte man neben sich und sogar unter sich Satelliten und Raumschiffe vorbeiflitzen sehen.

Es sei gesagt, dass das mit bloßem Auge kaum möglich ist, da sich die Objekte im Orbit mit etwa 7-8km/s bewegen. Und außerdem wird schon aus Sicherheitsgründen dafür gesorgt, dass sich Weltraumobjekte nicht zu nahe kommen und sich gegenseitig gefährden. Nicht nur, dass man Zusammenstöße natürlich verhindern will, da bei einem solchen wahrscheinlich nicht viel übrig bleiben würde, sondern auch deswegen, weil die jetzt verstreuten Einzelstücke gleich eine ganze Reihe anderer Satelliten gefährden könnten und im Schlimmsten Fall zu einer Kettenreaktion führen könnten, bei der der nahe Erdbereich (LEO) so sehr von Debree übersäht wird, dass eine weitere Nutzung des Weltraums zu gefährlich wird. Aus diesem Grund werden möglichst alle herumfliegenden Teile genauestens und mit mittlerweile sehr umständlichen Verfahren katalogisiert und analysiert um schon im Vorfeld vor möglichen Zusammenstößen gewarnt zu sein. Dank dieser Tatsache ist es möglich, kleinste Objekte von wenigen Zentimetern zu observieren und z.B. die ISS rechtzeitig auf eine sicherere Umlaufbahn ausweichen zu lassen. Aber nun wieder zurück zur Erdanziehungskraft.

Die Anziehungskraft ist also selbst in großer Höhe noch vorhanden. Das einzige, was einen Satelliten im Orbit hält ist nämlich seine horizontale Geschwindigkeit parallel zur Erdoberfläche.

Ich werde oft gefragt, wie es denn sein kann, dass in der Höhe der ISS die Anziehungskraft fast die gleiche ist, wie auf der Erdoberfläche, denn schließlich gäbe es ja die Astronauten, die in der ISS herumschweben können. Warum fallen die nicht auf den Boden der ISS?
→ Nun, das liegt daran, dass der gesamte ISS-Koloss, samt Gasinhalt und biologischer Masse als Einheit betrachtet werden kann, bei der sich alle Einzelteile mit der gleichen Geschwindigkeit parallel zueinander bewegen. Würde man die Gaswolke und die Metallteile um den Astronauten wegzaubern, dann würde er sich ungehindert mit der selben Geschwindigkeit, wie eben auch schon auf seiner Umlaufbahn weiterbewegen.

Wenn du jetzt einen Raketenstart betrachtest, dann wird dir auffallen, dass die Rakete gar nicht nur gerade nach oben fliegt, sondern in einigen Fällen schon eine Sekunde nach dem Start zur Seite kippt und damit ihr „Pitching Maneuver“ startet. *Das geschieht oft schon sehr früh, da die Rakete noch langsam genug ist, um die Flugrichtung in kurzer Zeit drastisch zu ändern.*

Aber warum macht die Rakete das?
→ Die Rakete wird, solange sie sich noch innerhalb der dichten Atmosphäre befindet, abgebremst. Deshalb müsste sie eigentlich zuerst nach oben fliegen, bis sie sich außerhalb der Atmosphäre befindet und sich dann hart nach „rechts“ drehen und solange Geschwindigkeit aufbauen, bis sie sich in einem Orbit befindet. Das geht in der Theorie vielleicht ganz ok, aber eine Geschwindigkeit von 8km/s parallel zur Erdoberfläche zu erreicht braucht lange. Da müsste man am Anfang

schon sehr hoch fliegen, damit man insgesamt genug Zeit hat, auch noch die nötige horzontale Geschwindigkeit zu erreichen. Wer mitdenkt wird natürlich feststellen, dass man so ja eine Menge Energie verschwendet, da die Geschwindigkeit nach oben ja nicht hilft, das Gefährt in einen Orbit zu setzen.
Darum hat man sich auch für eine kurvenförmige Flugbahn entschieden, die Steil beginnt und über die Zeit langsam abflacht, bis man schließlich nur noch in horizontaler Richtung Gas gibt. *Nochmal zur Wiederholung: Die Rakete fliegt nur nach oben um aus der abbremsenden Atmosphäre herauszukommen. Würde die Rakete vom Mond starten, so könnte sie sich die Energie nach oben eigentlich sparen und direkt zur Seite starten. Die einzige Einschränkung ist da vielleicht die Stärke der Düse (Thrust to Weight Ratio) und die Höhe der Berge.*

Auch interessant zu wissen: Besonders starke Raketenmotoren braucht es eigentlich nur auf der Erde. Einerseits wegen der Anziehungskraft, gegen die die Rakete am Anfang kämpfen muss und andererseits wegen der abbremsenden Atmosphäre.
Ein Satellit, der sich bereits in einem Orbit befindet, kann sich dafür seelenruhig Zeit lassen, da jede Energie, die er in die Geschwindigkeit in eine bestimmte Richtung steckt in gewisser Weise niemals wieder verloren geht. (Es besteht auch auf der Höhe der ISS eine gewisse Restatmosphäre. Deshalb muss sie alle paar Monate einen kleinen Schub in Flugrichtung geben um nicht irgendwann ungewollt in einen niedrigeren Orbit zu gelangen.)
Aus diesem Grund funktionieren auch „Ion thruster“ so gut bei Satelliten. Obwohl sie einen so schwachen Rückstoß erzeugen, dass man mit ihnen nicht einmal Modellbot bewegen könnte, sind sie auf der äußerst effizient bei der Umwandlung von chemischer Energie in kinetische Energie (Treibstoff wird umgewandelt in Bewegung). Da sie außerdem Monate lang Schub geben können, ist es möglich, mit ihnen über die Zeit hinweg eine viel höhere Geschwindigkeit zu erreichen, als es mit einer gängigen Raketendüse möglich gewesen wäre.

Auch interessant zu wissen: Bis zu einer Höhe von etwa 500Km ist die Atmosphäre auf der Erde noch soweit vorhanden, dass Satelliten relativ schnell abgebremst werden und über die Zeit automatisch Deorbiten. In dieser Höhe befanden sich in der Vergangenheit hauptsächlich Spy-Satelliten. 340km ist da schon das Minimum, damit ein Satellit einen stabilen Orbit aufrecht erhalten kann. Doch auch die ISS fliegt gerade mal in einer Höhe von etwa 400Km. Als die SpaceShuttles noch flogen, flog die ISS sogar noch tiefer. Die Restatmosphäre ist in dieser Höhe noch so hoch, dass die ISS einmal im Jahr wieder ihren Orbit anheben muss. Das ist sehr schön auf QUARTZ - every active satellite orbiting earth animiert worden. Wenn man dort die Animation startet, dann sieht man, dass die ISS immer zwischen 401-422km auf und ab pendelt.

Eine andere dreidimensionale Darstellung aller Erdnahen Teile auf stuffin.space. Wer möchte, kann dort sogar nach einzelnen Satelliten suchen und z.B. alle außer den GPS-Satelliten ausblenden.
Im Fall von SpaceX Starlink Internetsatelliten bleiben die Satelliten nur so lange im Orbit, wie sie noch Treibstoff haben. Das macht die Wartung solcher Satellitennetzwerke zwar etwas komplizierter, aber dafür hat man die Gewissheit, dass defekte Satelliten automatisch innerhalb von 5 Jahren deorbiten und daher kein Spacedebree hinterlassen. Anders als bei den meisten Satelliten mit Ionentriebwerk benutzt SpaceX anstatt des üblichen Xenon lieber Krypton als Reaktionsmasse. Andere verwendete Bezeichnungen für Ionentriebwerke sind unter anderem Hallantrieb oder Hall-Effekt-Antrieb/Hall-effect thruster (HET).

GPS-Satelliten sind übrigens im Medium Earth Orbit (MEO) angesiedelt.
Orbital altitudes of many significant satellites of earth
Comparison satellite navigation orbits

DRONE Solar System Model- How far is Planet 9? - Mark Rober
Watch This Guy Build a Massive Solar System in the Desert | Short Film Showcase - National Geographic
Sun and Earth to Scale

What Kerbal Space Program doesn't teach:
Reaction Wheels sind viel stärker als in der Realität.
RCS (Reaction-Control-Thruster) ist ebenfalls stärker als in der Realität.

(Keplersche Gesetze - Je höher der Orbit, desto geringer die Geschwindigkeit)

„Factor of safety“ ist für Raketen der Firma ULA zwischen 1.1 und 1.25, aber niemals darüber. Dieser Faktor ist dafür da um einen Sicherheitspuffer zur Verfügung zu haben, sollten Umstände eintreffen, die vorher nicht mitbedacht wurden. Bei Raketen ist der Sicherheitsfaktor sehr gering, da man die Rakete ja so leicht wie möglich belassen will und jeder Millimeter, den man z.B. der Dicke der Treibstofftanks hinzufügt, macht diesen zwar stabiler, aber fügt über die vielen Meter Fläche viele Kilo zusätzliches Gewicht hinzu, was stärkere Engines und mehr nötiger Treibstoff bedeuten könnte. Im Gegensatz dazu hat ein Traktor einen sehr viel höheren Sicherheitsfaktor, denn bei diesem sind Stabilität und Kraft sehr viel mehr Wert als Eigengewicht und Geschwindigkeit. Siehe auch HOW ROCKETS ARE MADE (Rocket Factory Tour - United Launch Alliance) - Smarter Every Day 231.

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Ort, Datum Unterschrift